Your Answers:

1 1 / 1 point

If an unauthorized user gets read access to my files, the security property of integrity is violated.

- ○ True
- ✓ ● False

Feedback

Based on your answer

The statement is FALSE. Reasoning: It is a violation of confidentiality. Since the file is not modified by read access, integrity is preserved.

2 1 / 1 point

If Alice and Bob want to privately exchange messages, they need the items listed below. Once those are in place, they can send and receive messages from/to each other without risking that Eve eavesdrops or Mallory messes with the messages.

- the plaintext message
- an encryption algorithm
- a decryption algorithm

- ○ True
- ✓ ● False

Feedback

Based on your answer

The statement is false. Reasoning: They need to have keys, either a shared secret key if they use symmetric cryptography or a key pair (both a public and a corresponding private key) each. They need a key-generation algorithm to create those and a way of communicating the needed keys (the shared symmetric key or each other's public keys, respectively).

3 1 / 1 point

The reason for individual salts for hashing passwords is to make easy-to-guess passwords more difficult to guess.

- ○ True
- ✓ ● False

Feedback

Based on your answer

Yes, indeed. The salts are not secret, so they don't make guessing a particular password more difficult. So they don't compensate for bad passwords.

The purpose of a salt is to make two hashes different even if the two passwords are the same (because the salts are randomly chosen, so they will differ). This makes passwords harder to guess, but that's because the attacker must guess each password individually, as opposed to guessing all at once.

Salts render rainbow tables unusable, but this affects the difficulty of all passwords --- not just easy-to-guess passwords.

4 1 / 1 point

Authentication has two parts: claim that an attribute fulfils a certain predicate, convince a verifier that is indeed true.

- ✓ ◉ True
- ○ False

Feedback

Based on your answer

We must first establish with the verifier what to authenticate. Second, we must prove that the statement holds.

For instance, we can claim that our age at least 18 years. We can prove it by showing an ID card, which is someone the verifier trusts certifying our birthday; from which our age can be reliably computed.

5 1 / 1 point

For access control to be meaningful, the subject must be authenticated.

- ✓ ◉ True
- ○ False

Feedback

Based on your answer

Yes. If we're not sure the subject has the required attributes, anyone can access and possibly violate the access policy.

6 1 / 1 point

With discretionary access control, a subject with access to an object (e.g., resource, file) can decide to grant access rights to the next user.

- ○ True
- ✓ ◉ False

Feedback

Based on your answer

This depends on the policy. The user can grant access for another user if they have that access right. E.g. in the traditional UNIX file system, the owner has that right, but no

one else.

So in general, the statement is false.

7 1 / 1 point

Social engineers use strategies that exploit even desirable human traits, such as the tendency to reciprocate for a favor, that are independent of intelligence.

- ✓ ● True
- ○ False

Feedback

Based on your answer

The statement is TRUE. Reasoning: social engineers aim to put the target into situations that reduce the target's likelihood and ability to systematically think through a request before responding, for example by creating stress, time pressure, fear of repercussions, or altercasting dynamics to push the target into a role. When systematic thinking is preempted, socialized, habitual, and reflexive responses prevail.

8 1 / 1 point

Stateless packet-filtering firewalls filter out any malicious data from packets.

- ○ True
- ✓ ● False

Feedback

Based on your answer

The statement is FALSE. Reasoning: They only look at the header of a packet and decide by the rules if it should be dropped (removed entirely) or passed on to its intended recipient.

9 1 / 1 point

Deep packet-inspection firewalls can decrypt encrypted connections to look at the data in the payload.

- ○ True
- ✓ ● False

Feedback

Based on your answer

Correct. If a connection is end-to-end encrypted, the DPI firewall cannot look at the data. Then it can only consider meta-data, the fact that the connection is encrypted and so on.

10 1 / 1 point

Signature-based intrusion detection can only detect intrusions that are already identified as such in the IDS.

- ✓ ● True
- ○ False

Feedback

Based on your answer

Signature-based detection requires that we know what an intrusion looks like. A signature is a pattern that we can recognize as malicious. Thus we must know it in advance.

11 1 / 1 point

Your company bought the best intrusion detection system on the market. It has only 0.01% false positives. However, your company is overwhelmed with false alarms, so the manufacturer must have lied about the false-positive rate in their marketing material.

- ○ True
- ✓ ● False

Feedback

Based on your answer

Yes, indeed. This can happen even with a very low false positive rate like that. If the number of true positive cases are much smaller than true negative cases, then those 0.01% of false positives can grow very large.

12 1 / 1 point

A motivation to build and control a botnet is to use the botnet's resources to mine bitcoins.

- ✓ ● True
- ○ False

Feedback

Based on your answer

The answer is TRUE, in this case the financial gain is a motivation to control a large number of computational resources, which mine bitcoin for the adversary.

13 1 / 1 point

Malware cannot infect computers that are not network connected.

- ○ True

○ True

✓ ◉ False

Feedback

Based on your answer

The answer is False. A malware may compromise a removable storage, like a USB-stick. When the USB stick is inserted in a computer and a program accesses a file on the USB stick, the file may corrupt the program due to a vulnerability.

14 1 / 1 point

In case of an ICMP flood, the victim can counter the attack by dropping (ignoring) all packets that are sent by the attacker.

○ True

✓ ◉ False

Feedback

Based on your answer

The answer is FALSE. The packet should be dropped before the congested link. Even if the victim drops received packets its network link will be congested by the adversary traffic.

15 1 / 1 point

The return pointer is usually stored in the heap. For this reason, an overflow of a buffer that is allocated in the heap may corrupt the control flow of the software.

○ True

✓ ◉ False

Feedback

Based on your answer

The statement is FALSE. The return pointer is not stored in the heap, it is stored in the stack. However, an overflow of a buffer in the heap can lead to control flow corruption if function pointers are stored in the heap.

16 1 / 1 point

Diversification techniques can be used to mitigate buffer overflows. I.e. they hinder the adversary's ability of exploiting them.

✓ ◉ True

○ False

Feedback

Based on your answer

The statement is TRUE. For instance, if we diversify the position of buffers in memory it can be hard for the attacker to know how many bytes they should write in a buffer to override another variable and obtain a specific effect.

17 1 / 1 point

A methodology to prevent cache side channels is to guarantee that the program conditionals and memory accesses do not depend on secret data.

- ✓ ● True
- ○ False

Feedback

Based on your answer

The answer is TRUE. This is also called constant time programming. Since the state of the cache depends on the executed instructions and the accessed addresses, if a program performs exactly the same branches and the same memory accesses independently of secret data, then the state of the cache is independent of confidential information

18 1 / 1 point

The notion of privacy goes beyond confidentiality and secrecy and includes wider aspects such as control and context.

- ✓ ● True
- ○ False

Feedback

Based on your answer

The statement is TRUE. Confidentiality and secrecy are not enough to prevent inferences. Without exercising control over what is disclosed to whom, directly or indirectly via inferences, and taking into account what is appropriate for a specific context, privacy breaches (involuntary disclosure of information) can happen even if technically the confidentiality of data is preserved.

19 18 points possible

Upload the file containing your reasoning for the answers to all questions.

(no answer)

Waiting for grade